

Seyhak Manith Chhun

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Electrical Engineering Student (Co-op) | Power/Energy Systems · Automation · Software Tools

Second-year Electrical Engineering Co-op student with a Computer Science DEC, interested in power/energy systems, embedded electronics, automation, and hardware-software integration. Project experience includes Arduino data acquisition, Python automation, LTspice circuit simulation, and Unity/C# real-time systems.

Education

Bachelor of Engineering, Electrical Engineering (Co-op) — Concordia University, Montréal, QC | Expected 2029

Diploma of College Studies (DEC), Computer Science, Pre-University Program — Collège Lionel-Groulx, Sainte-Thérèse, QC | 2025

Technical Skills

Programming: Python, C++, C#, JavaScript, HTML/CSS, Git, XML, SQL, Object-Oriented Programming(OOP)

Frameworks/Tools: Node.js, React, RestAPIs, API Integration, web scraping, FastAPI, Pydantic, Excel, LTspice, Unity, AI, Arduino, RSS

Electrical: Circuit analysis, KCL/KVL, Thevenin/Norton, AC circuits, phasors, op-amps, digital logic (logic gates using ICs)

Data/Automation: CSV/Excel processing, data logging, engineering calculations, basic plotting

Languages: English (**Fluent**), French (**Fluent**)

Projects

Mini Wind Energy Harvester — In Progress

Arduino · Python/Excel · Electrical Measurements · Electromagnetism

- Building a fan-driven wind energy test setup using magnets and coils to generate AC electrical output and implementing AC-to-DC conversion (rectification) to obtain usable DC power.
- Using a digital multimeter (DMM) for validation and Arduino for analog data acquisition, with Python/Excel for logging, calculations, and performance plots.
- Analyzing how airflow, magnetic induction, and load resistance affect generated power, measurement accuracy, and overall system performance.

Dream Game — Unity Game Development Project (Team Project)

Unity · C# · Vector IK · Animation · Collision Systems

- Collaborated with a team to develop a Unity game prototype, contributing gameplay systems and technical implementation.
- Implemented a vector-based inverse kinematics arm simulation to control procedural limb movement.
- Developed animation and collision-based combat mechanics, including hit detection, attack interactions, and battle behavior.

SigFeed — Personalized News Intelligence Platform — In Progress

Python · APIs · Web Scraping · Automation · AI

- Building a Python-based data pipeline that ingests online news/articles through APIs and web scraping from selected sources.
- Developing an AI-assisted filtering and scoring system to interpret news/articles, prioritize high-signal content, and reduce low-value items.
- Implementing automated workflows for recurring data collection, storage, and personalized delivery.

Electrical Engineering Lab & Simulation Work

LTspice · Circuit Analysis · Electrical Measurements

- Simulated and analyzed circuits involving resistors, capacitors, inductors, op-amps, AC sources, and transient behavior using LTspice.
- Applied KCL/KVL, Thevenin/Norton equivalents, impedance analysis, and phasors to compare theory and simulation.
- Interpreted voltage, current, gain, frequency response, and transient behavior in electrical systems.

Work Experience

Sales Associate — Ecko Unltd, Rosemère, QC | 2022–2023

- Delivered customer service and sales support in a fast-paced retail environment.
- Developed strong communication, teamwork, and problem-solving skills through direct customer interactions.
- Managed multiple tasks while maintaining attention to detail, organization, and professionalism.